

Steady Flow in the Transition Length of a Straight Tube¹

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By means of a linearizing approximation, the Navier-Stokes equations are solved for the case of steady flow in the transition length of a straight tube. The family of velocity profiles is defined by Bessel functions, and the parameter of this family is tabulated against the axial coordinate in a dimensionless form. Hence, the length of transition is obtained. The curves give a comparison of the author's calculations of the velocity field with those of other investigators,^{3,4,5} and with the experimental data of Nikuradse.⁶ The pressure function is derived from the computed velocity field by means of the energy equation, and the pressure drop in the transition length is defined by a dimensionless constant m , which is computed to be 2.28. A discussion of this constant is given in the conclusions.

NOTATION

x, y, z = rectangular coordinates of a point. The center line of the tube is on the z axis, with the inlet at $z = 0$
 r, θ, z = cylindrical co-ordinates of the point x, y, z
 ω = cross-sectional area of the tube
 a = radius of the tube
 $q = r/a$ = dimensionless radial co-ordinate
 u, v, w = x, y, z components of the velocity field
 w_0 = velocity on the center line of the tube
 c = mean velocity in the tube
 $\lambda = w/c$ = dimensionless axial velocity
 $\lambda_0 = w_0/c$ = dimensionless velocity on the center line of the tube
 ρ = mass-density of the liquid
 ν = kinematic coefficient of viscosity
 $\mu = \rho\nu$ coefficient of viscosity
 $R = ca/\nu$ = Reynolds number
 $\sigma = z/aR$ = dimensionless axial co-ordinate
 σ' = value of σ corresponding to $\lambda_0 = 1.98$
 g_3 = the z component of the acceleration of gravity

p = static pressure at the point x, y, z

p_0 = static pressure at the inlet section $z = 0$

Δp = pressure drop between the still liquid in the supply reservoir and the section z

m = dimensionless constant which defines the pressure drop in the transition length

α, β = certain parameters which are functions of z alone

$\gamma = \beta\alpha$ = dimensionless parameter

$I_n(x) = i^{-n}J_n(ix)$, where $i = \sqrt{-1}$, and J_n is the n th-order Bessel function.⁷ Throughout this paper the argument of the I functions is γ if not otherwise indicated

THE FAMILY OF VELOCITY PROFILES

The Navier-Stokes equation for w is

$$dw/dt = g_3 - (1/\rho)(\partial p/\partial z) + \nu(w_{xx} + w_{yy} + w_{zz}) \dots [1]$$

The following three approximations, which are common to the theories of Boussinesq, Schiller, and Atkinson-Goldstein, will be employed in the present theory:

1 Over the inlet section $z = 0$, the velocity component w is constant and equal to the mean velocity c . Nikuradse's⁶ test curves thoroughly justify this assumption for the case of tubes which are provided with short bell-shaped mouths.

2 The pressure gradient $-\partial p/\partial z$ is a function of z alone.

3 The term νw_{zz} may be neglected in Equation [1]. This is tantamount to the assumption that the derivative of the normal stress $\partial\sigma_z/\partial z$ is the same as the pressure gradient $-\partial p/\partial z$.

In addition to these assumptions, there is introduced the fourth condition

$$dw/dt = \nu\beta^2 w \dots \dots \dots [2]$$

where β is a function of z alone. The purpose of this approximation is to reduce Equation [1] to a linear form. It can be partially justified on theoretical grounds. In the first place, Equation [2] is satisfied by $\beta = 0$ in the permanent regime beyond the transition length. Also, this equation is satisfied, irrespective of the value of β , at all points on the wall of the tube, since w and dw/dt both vanish there. Furthermore, if it is assumed with Schiller⁴ and Atkinson-Goldstein⁵ that the laminar state is attained by the development of a boundary layer in the transition length, then the conclusion is reached that Equation [2] is satisfied at all points in the undisturbed central core of fluid. This conclusion emerges from the fact that the derivatives w_x and w_y vanish over the flat central portions of the velocity profiles, and accordingly the acceleration $dw/dt = uw_x + vw_y + ww_z$ reduces to $dw/dt = ww_z$. Since w is a function of z alone in the central core, w_z is likewise a function of z alone in that region, and it follows that Equation [2] is there satisfied by $\nu\beta^2 = w_z$. Finally, it may be concluded from this that Equation [2] is satisfied at all points of the inlet section $z = 0$, since the depth of the boundary layer there reduces to zero. Thus, Equation [2] is justified, except for those points which lie within the boundary layer. Of

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³ "Hydrodynamique," by J. Boussinesq, *Comptes Rendus*, vol. 110, 1890, pp. 1160, 1238; vol. 113, 1891, pp. 9, 49.

⁴ "Die Entwicklung der laminaren Geschwindigkeitsverteilung und ihre Bedeutung für Zähigkeitsmessungen," by L. Schiller, *Zeitschrift für angewandte Mathematik und Mechanik*, vol. 2, 1922, p. 96.

⁵ "Modern Developments in Fluid Dynamics," edited by S. Goldstein, Oxford University Press, London, England, and New York, N. Y., 1938, vol. 1, Atkinson-Goldstein, p. 304.

⁶ "Applied Hydro- and Aeromechanics," by L. Prandtl and O. G. Tietjens, McGraw-Hill Book Co., Inc., New York, N. Y., first edition, 1934. Nikuradse's curves are reproduced from Fig. 13, p. 27. Seemingly a more detailed account of Nikuradse's tests has not been published.

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course, in the latter part of the transition length, a true boundary layer does not exist, since the central parts of the velocity profiles are appreciably rounded. However, in that region, the flow approaches the laminar state for which Equation [2] remains valid.

With the foregoing assumptions, Equation [1] reduces to the form

$$w_{xx} + w_{yy} - \beta^2 w = \alpha \dots \dots \dots [3]$$

where α and β are functions of z alone. This equation will apply for a cylindrical tube of any cross section, but the present discussion is restricted to the circular cross section.

A particular solution of Equation [3] is

$$w = -\alpha/\beta^2 \dots \dots \dots [4]$$

It then suffices to solve the complementary equation

$$V_{xx} + V_{yy} - \beta^2 V = 0 \dots \dots \dots [5]$$

In cylindrical co-ordinates, this reduces to Bessel's equation of order zero

$$V_{rr} + \frac{1}{r} V_r - \beta^2 V = 0 \dots \dots \dots [6]$$

The only solution which is finite for $r = 0$ is $V = KI_0(\beta r)$. Hence

$$w = KI_0(\beta r) - \alpha/\beta^2 \dots \dots \dots [7]$$

The quantities α and K may be eliminated from this equation by means of the boundary condition $w = 0$, and the continuity equation

$$\int_0^a r w dr = (1/2) a^2 c \dots \dots \dots [8]$$

Introducing the dimensionless notation $q = r/a$, $\lambda = w/c$, $\gamma = \beta a$, gives

$$\lambda = [I_0(\gamma) - I_0(\gamma q)]/I_2(\gamma) \dots \dots \dots [9]$$

This is the equation of the family of velocity profiles. It remains to express the parameter γ of this family of curves as a function of z .

THE FUNCTION $\gamma(z)$

The problem of expressing the parameter of a known family of velocity profiles as a function of the axial co-ordinate is treated in the paper by Schiller.⁴ In fact, Schiller assumes the velocity profiles to be straight-line segments terminated by parabolic arcs, and he obtains the parameter of this family as a function of z by applying the momentum equation to the whole body of fluid, and Bernoulli's equation to the central core. Essentially the same treatment is given here but, instead of using the simple Bernoulli equation, the two viscosity terms νw_{xx} and νw_{yy} in the Navier-Stokes equation are retained, since they are important for the latter part of the transition length. These terms are irrelevant in Schiller's treatment, since, by his hypotheses, they vanish.

The momentum equation, which can be obtained by integrating the Navier-Stokes equation over the cross section, is

$$2\nu \int_0^a (r w_{rr} + w_r) dr + a^2 g_3 - \frac{a^2}{\rho} \frac{dp}{dz} = 2 \frac{d}{dz} \int_0^a w^2 r dr \dots [10]$$

By means of Equation [9], the first integral may be evaluated, giving

$$-\frac{2\nu c \gamma I_1(\gamma)}{I_2(\gamma)} + a^2 \left(g_3 - \frac{1}{\rho} \frac{dp}{dz} \right) = 2 \frac{d}{dz} \int_0^a w^2 r dr \dots [11]$$

The term $g_3 - (1/\rho) dp/dz$ may be eliminated by means of the Navier-Stokes equation⁵

$$g_3 - \frac{1}{\rho} \frac{dp}{dz} = w_0 \frac{dw_0}{dz} - \nu (w_{rr} + \frac{1}{r} w_r)_0 \dots \dots \dots [12]$$

where the subscript zero refers to the central axis. Equations [9] and [12] yield

$$g_3 - \frac{1}{\rho} \frac{dp}{dz} = w_0 \frac{dw_0}{dz} + \frac{\nu c \gamma^2}{a^2 I_2(\gamma)} \dots \dots \dots [13]$$

Substituting this relation in Equation [11] gives an equation which may be expressed in the following dimensionless form

$$\frac{2\gamma I_1(\gamma) - \gamma^2}{I_2(\gamma)} = \frac{d}{d\sigma} \int_0^1 (\lambda_0^2 - 2\lambda^2) q dq \dots \dots \dots [14]$$

Also, by Equation [9]

$$\int_0^1 (\lambda_0^2 - 2\lambda^2) q dq = \frac{(I_0 - 1)^2 - 4I_0 I_2 + 2I_1^2}{2I_2^2} \dots \dots [15]$$

in which the argument of the I 's is γ . It is convenient to use the abbreviations

$$\begin{aligned} f(\gamma) &= [4I_0 I_2 - (I_0 - 1)^2 - 2I_1^2]/2I_2^2 \\ g(\gamma) &= I_2/(2\gamma I_1 - \gamma^2) \end{aligned} \dots \dots [16]$$

Equation [14] then may be expressed

$$1/g(\gamma) = -f'(\gamma) d\gamma/d\sigma \dots \dots \dots [17]$$

This is a differential equation which determines γ as a function of σ .

By means of the asymptotic formula

$$I_n(x) = \frac{e^x}{(2\pi x)^{1/2}} [1 + \epsilon_n(x)], \quad \lim_{x \rightarrow \infty} \epsilon_n(x) = 0 \dots \dots [18]$$

there results from Equation [9] $\lim_{\gamma \rightarrow \infty} \lambda = 1$. Since $\lambda = 1$ at the inlet section $z = 0$, then $\lim_{\gamma \rightarrow \infty} z(\gamma) = 0$. Hence, Equation [17] yields

$$\sigma = \int_{\gamma}^{\infty} g(\gamma) f'(\gamma) d\gamma = \int_{\gamma}^{\infty} g(\gamma) df(\gamma) \dots \dots [19]$$

For the numerical integration of this equation, make the change of variables

$$X = f(\gamma), \quad Y = g(\gamma) \dots \dots \dots [20]$$

By the power series and the asymptotic series for the I functions the following limits are established

$$\begin{aligned} X &\rightarrow 1/2, Y \rightarrow 0 \quad \text{as } \gamma \rightarrow \infty \\ X &\rightarrow -2/3, Y \rightarrow \infty \quad \text{as } \gamma \rightarrow 0 \end{aligned} \dots \dots [21]$$

Equation [19] may then be expressed

$$\sigma = \int_X^{1/2} Y dX \dots \dots \dots [22]$$

Equation [20] may be regarded as the parametric equations of a curve, and Equation [22] shows that the function $\sigma(\gamma)$ is represented by the area under this curve in the interval $(X, 1/2)$. By assigning a set of values to γ , a table of points on the curve may be computed, which gives, by numerical integration, the corresponding values of σ . The values of σ in Table 1 are computed

⁵ Schiller here employed the Bernoulli equation. This is obtained from Equation [12] by dropping the viscosity terms.

TABLE 1

γ	100.00	30.00	20.00	16.00	14.50	13.00	12.00
σ	0.000026	0.000334	0.000820	0.00137	0.00172	0.00222	0.00269
λ_0	1.0203	1.0701	1.1079	1.1377	1.1535	1.1734	1.1898
γ	11.00	10.00	9.00	8.00	7.00	6.50	6.00
σ	0.00332	0.00418	0.00541	0.00722	0.00997	0.01188	0.0143
λ_0	1.2095	1.2337	1.2638	1.3021	1.3514	1.3814	1.4157
γ	5.50	5.00	4.50	4.00	3.50	3.25	3.00
σ	0.0174	0.0214	0.0267	0.0335	0.0426	0.0483	0.0549
λ_0	1.4547	1.4989	1.5488	1.6041	1.6645	1.6961	1.7285
γ	2.75	2.50	2.25	2.00	1.85	1.70	1.55
σ	0.0625	0.0715	0.0821	0.0947	0.1034	0.1132	0.1241
λ_0	1.7612	1.7939	1.8261	1.8573	1.8753	1.8926	1.9090
γ	1.40	1.30	1.20	1.10	1.00	0.90	0.80
σ	0.1365	0.1459	0.1560	0.1671	0.1795	0.1934	0.2091
λ_0	1.9245	1.9342	1.9434	1.9520	1.9600	1.9673	1.9740
γ	0.70	0.60	0.50	0.40	0.00		
σ	0.2270	0.2479	0.2730	0.3040			
λ_0	1.9800	1.9852	1.9897	1.9934	2.0000		

in this way. The values of λ_0 are computed directly from Equation [9].

This table, with Equation [9], completely defines the velocity field in the transition length. The value $\sigma = \sigma'$, corresponding to $\lambda_0 = 1.98$, was called by Boussinesq the "length of transition." From Table 1 it follows $\sigma' = 0.23$.

THE PRESSURE DROP

In the transition length, the pressure gradient $-dp/dz$ is higher than in a region of laminar flow, not only because of the increased frictional loss, but also because the kinetic energy of the liquid increases as it passes downstream. To evaluate the pressure drop, consider the energy equation. The power which is dissipated by fluid friction in the interval $z = 0$ to $z = z$ is⁹

$$E = \mu \int_0^z dz \int_{\omega} [2(u_x^2 + v_y^2 + w_z^2) + (w_y + v_z)^2 + (u_z + w_x)^2 + (v_z + u_y)^2] d\omega \dots [23]$$

It is assumed that the lateral velocity components u and v may be neglected in this equation. Then the continuity equation $u_z + v_y + w_x = 0$ shows that consistently the term w_x may be neglected, so that the preceding equation reduces to

$$E = \mu \int_0^z dz \int_{\omega} (w_x^2 + w_y^2) d\omega \dots [24]$$

By a theorem of Helmholtz,¹⁰ the laminar velocity distribution furnishes a minimum to the integral [24], and accordingly the friction loss is always higher in the transition length than it is in the permanent regime further downstream.

During a time interval dt , the net work of pressure on the body of fluid in the interval $z = 0$ to $z = z$ is $c\omega(p_0 - p)dt$. The work of friction on this body of fluid is $-Edt$, where E is given by Equation [24]. It is assumed that the tube is horizontal, so that the work of body force is zero. Then, equating the work of all forces to the change in kinetic energy gives the energy equation

$$c\omega(p_0 - p) = \mu \int_0^z dz \int_{\omega} (w_x^2 + w_y^2) d\omega + (1/2)\rho \int_{\omega} (w^3 - c^3) d\omega \dots [25]$$

For the circular cross section, this may be expressed

$$\frac{p_0 - p}{(1/2)\rho c^2} = 2 \int_0^1 (\lambda^3 - 1)q dq + 4 \int_0^{\sigma} d\sigma \int_0^1 \lambda_q^2 q dq \dots [26]$$

By Bernoulli's equation, the pressure drop that occurs in the mouth of the tube is $(1/2)\rho c^2$. Accordingly, the total drop Δp be-

⁹ "Hydrodynamics," by Horace Lamb, Cambridge University Press, London, England, sixth edition, 1932, par. 329, p. 580.

¹⁰ See footnote, ⁹ par. 344, pp. 617, 618.

TABLE 2

γ	σ	$\int_0^1 (\lambda^3 - 1)q dq$	$\int_0^{\sigma} \phi(\gamma) d\sigma$	$\Delta p / (1/2)\rho c^2$
20.0	0.00082	0.0677	0.0384	1.2122
11.0	0.00332	0.1112	0.0821	1.3866
8.0	0.00722	0.1540	0.1326	1.5732
6.0	0.0143	0.2055	0.2095	1.8300
5.0	0.0214	0.2433	0.2788	2.0442
4.0	0.0335	0.2923	0.3884	2.3614
3.0	0.0549	0.3531	0.5706	2.8474
2.5	0.0715	0.3867	0.7082	3.1898
2.0	0.0947	0.4210	0.8967	3.6354
1.4	0.1365	0.4588	1.2332	4.3840
1.0	0.1795	0.4796	1.5777	5.1146
0.6	0.2479	0.4912	2.1251	6.2326
0.4	0.3040	0.5000	2.5739	7.1478

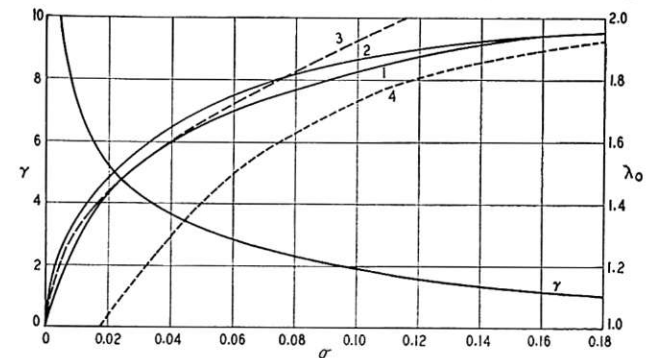


FIG. 1 CENTER-LINE VELOCITIES AS COMPUTED BY VARIOUS THEORIES

Curve 1—Observed by Nikuradse
Curve 2—Plotted from Table 2
Curve 3—Calculated by Schiller's formula
Curve 4—Calculated by Boussinesq's formula

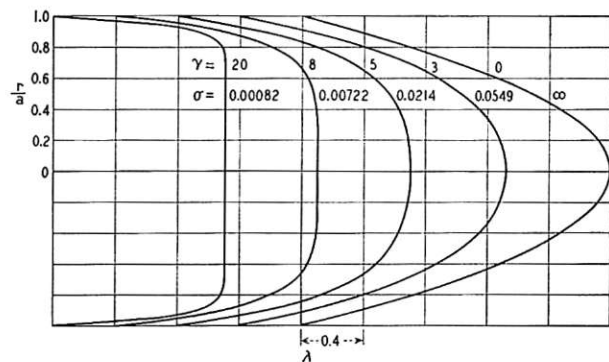


FIG. 2 VELOCITY PROFILES
(Computed by Equation [9] and Table 1.)

tween the supply reservoir and the section z is $(1/2)\rho c^2 + (p_0 - p)$. Equations [9] and [26] then give

$$\frac{\Delta p}{(1/2)\rho c^2} = 1 + 2 \int_0^1 (\lambda^3 - 1)q dq + 2 \int_0^{\sigma} \frac{\gamma^2}{I_2^2} (I_1^2 - I_0 I_2) d\sigma \dots [27]$$

It is convenient to denote the integrand $(\gamma^2/I_2^2)(I_1^2 - I_0I_2)$ by $\phi(\gamma)$. The evaluation of the integral of this function involves a difficulty, due to the fact that the integrand becomes infinite when $\sigma \rightarrow 0$. However, the relation $\lim_{\gamma \rightarrow \infty} \gamma^2\sigma = 1/4$ (which may be derived from Equation [19]), and the asymptotic formula

$$\phi(\gamma) \sim \gamma + 3 + \frac{5}{\gamma} + \frac{7}{\gamma^2} + \dots [28]$$

$$\text{gives} \quad \int_0^{0.0426} \phi(\gamma) d\sigma = 0.00518 \dots [29]$$

($\sigma = 0.000026$ corresponds to $\gamma = 100$).

Hence

$$\int_0^\sigma \phi(\gamma) d\sigma = 0.00518 + \int_{0.0426}^\sigma \phi(\gamma) d\sigma \dots [30]$$

Since $\int_{0.0426}^\sigma \phi(\gamma) d\sigma$ is a proper integral, it may be evaluated by numerical methods. Also, the integral $\int_0^1 (\lambda^3 - 1)q dq$ may be calculated by Simpson's rule from Equation [9]. With Equation [27], Table 2 is then obtained.

From this table we may plot a curve of $\Delta p / (1/2)\rho c^2$ against σ . This curve determines the pressure drop Δp as a function of z . The slope of the curve is

$$-\frac{dp/d\sigma}{(1/2)\rho c^2} = -\frac{a^2}{(1/2)\rho v c} \frac{dp}{dz} \dots [31]$$

In the region of laminar flow, $dp/dz = -8c\rho v/a^2$. Consequently, the curve is asymptotic to a line with slope 16.

If a value σ_1 is chosen somewhat greater than the length of transition σ' , and if $\sigma > \sigma_1$, Equation [27] may be expressed in the form

$$\frac{\Delta p}{(1/2)\rho c^2} = 1 + 2 \int_0^1 (\lambda^3 - 1)q dq + 2 \int_0^{\sigma_1} \phi(\gamma) d\sigma + 2 \int_{\sigma_1}^\sigma \phi(\gamma) d\sigma \dots [32]$$

But, for $\sigma > \sigma_1$, $\lambda = 2(1 - q^2)$ (Poiseuille flow), and consequently $\int_0^1 (\lambda^3 - 1)q dq = 1/2$. Also, from the power series for the I

functions, $\gamma^2/I_2 \rightarrow 8$ and $(I_1^2/I_2) - I_0 \rightarrow 1$ as $\gamma \rightarrow 0$. Since $\gamma \rightarrow 0$ as $z \rightarrow \infty$, it follows

$$\int_{\sigma_1}^\sigma \phi(\gamma) d\sigma = 8(\sigma - \sigma_1) \dots [33]$$

Hence, setting $\int_0^{\sigma_1} \phi(\gamma) d\sigma = K$, gives

$$\Delta p / (1/2)\rho c^2 = 16\sigma + 2K + 2 - 16\sigma_1 \dots [34]$$

Setting $2K + 2 - 16\sigma_1 = m$, this equation may be expressed in the form

$$\Delta p = \frac{8\mu cz}{a^2} + (1/2)m\rho c^2 \dots [35]$$

This is a fundamental equation in the theory of tube-type viscosimeters. If $\sigma_1 = 0.304$, Table 2 shows $K = 2.574$. Hence

$$m = 2.28 \dots [36]$$

CONCLUSIONS

The velocity field in the transition length is completely defined by Equation [9] and Table 1. The curves of Fig. 1 give a comparison of the center-line velocities as computed by this theory, and by the theories of Boussinesq³ and Schiller,⁴ and as observed by Nikuradse.⁶ From Nikuradse's curve, the computed length of transition $\sigma' = 0.23$, appears to be in close agreement with experiment. A few of the computed velocity profiles are shown in Fig. 2.

The pressure function in the transition length is defined by Table 2. However, if the coordinate z is greater than the length of transition, the pressure drop Δp is given by Equation [35], in which, according to the present theory, $m = 2.28$. The most suitable value for this constant has not been well established by experiment. Schiller computed $m = 2.16$, but he says that this value seems to be too low. In four runs, he obtained the experimental values 2.115, 2.35, 2.36, and 2.45—average 2.32.¹¹ Other experimental results have been given by Riemann,¹² who obtained values ranging from 2.220 to 2.268—average 2.248 $\pm$ 0.012. Accordingly, he recommends for viscosimetry practice the value 2.24, which was computed by Boussinesq. Atkinson and Goldstein⁵ computed $m = 2.41$, but this value appears to be high.

¹¹ "Handbuch der Experimentalphysik, IV, 4 Teil, Hydro- und Aerodynamik," by Ludwig Schiller, Akademische Verlagsgesellschaft M. B. H., Leipzig, Germany, 1932, p. 57.

¹² "The Value of the Hagenbach Factor in the Determination of Viscosity by the Efflux Method," by William Riemann, 3rd, *Journal of the American Chemical Society*, vol. 50, 1928, p. 46.